

Block 4 – Cloud Computing

S2: Cloud Computing & Cloud Architecture

Arquitectura de Xarxes

Universitat Pompeu Fabra



- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture
- 5 Cloud Network Services
- 6 The Cloud Trade-Off
- 7 Session Summary

Outline

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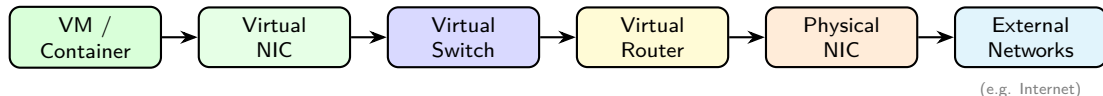
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- 3 Design a cloud virtual network with subnets, route tables, and security controls
- 4 Identify essential cloud network services: load balancers, proxies, and VPNs

Recap: What We Virtualized (Session 1)



- **Compute:** VMs share physical hardware via hypervisors; containers share the host kernel directly
- **Networking:** Virtual NICs, Switches, and Routers replicate the physical stack in software
- **Isolation:** VLANs and overlay networks separate tenants on shared infrastructure

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You own the hardware	Provider owns the hardware

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Meet the Scenario: A Startup from Zero

Running example for this session

To understand concepts in the cloud, we will use "**CloudBite**". This is a 3-person startup that we just created and received seed funding. We need to deploy a food-delivery web app for thousands of users **by next month**.

We have laptops, code, and zero infrastructure.

Discussion: From Virtualization to Cloud

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Think about:

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*"A model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources that can be rapidly provisioned and released with **minimal management effort**."*

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- **Minimal effort**: you click a button (or call an API); the provider handles the rest

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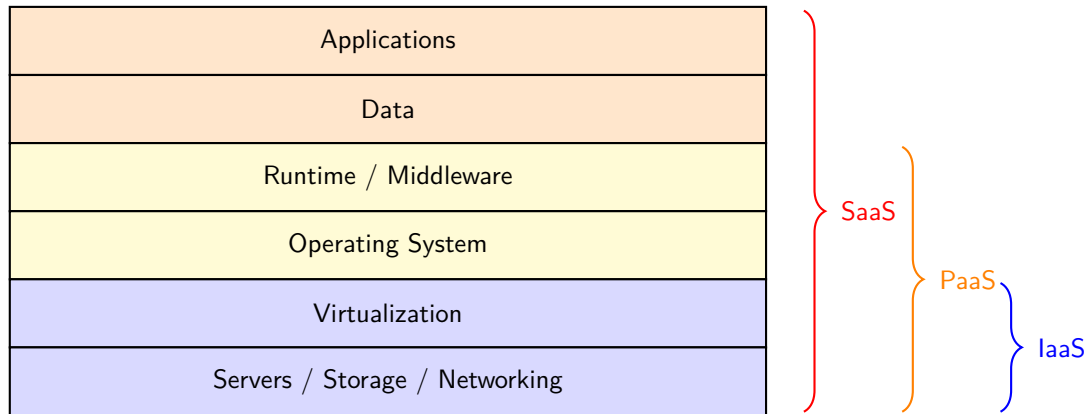
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Rapid elasticity	Scale up/down automatically	Black Friday: 10× more instances
Measured service	Pay only for what you use	Billed per CPU-hour, GB stored

All five must be present to qualify as “cloud computing”

Cloud Service Models – The Stack



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SaaS (e.g. Google Workspace, Microsoft 365, Slack, Zoom):

- Provider gives you a complete application. You manage your data only
- “I need email for 500 employees. I do not want to run a mail server”

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
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Hardware	Provider	Provider	Provider
Network Services (LB, Proxy, VPN)	Provider	Provider	Provider

IaaS → SaaS: **less control, less responsibility.**

Major providers (AWS, Azure, GCP) offer all three models.

Discussion: Cloud Fundamentals

Case A: University

A large university wants a web tool for students to submit assignments. The IT department has 10 staff. Student data must stay within EU jurisdiction. Budget is fixed yearly.

Which model? Why?

- Control over data location needed
- Existing IT team can manage servers
- Ready-made products exist (Canvas, Moodle)
- No need to scale suddenly

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Case B: CloudBite

We are 3 engineers. We need to run a web app, store orders, and send email notifications. We must ship in 4 weeks and cannot afford to manage servers.

Which model? Why?

- No ops bandwidth: avoid IaaS if possible
- Web app: PaaS (just deploy the code)
- Email: SaaS (Mailchimp, SendGrid)
- Database: PaaS managed DB, or IaaS if we need fine-grained control

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Let us look at the platforms available.

Putting It All Together

What we have seen (Session 1)

- Virtual Switches and Virtual Routers
- Virtual NICs and VLANs
- Hypervisors (KVM, VMware) and containers

These are the **building blocks** of any virtual network; they do not run in isolation.

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Platforms that manage them

- **Cloud platforms:** orchestrate VMs, storage, networking at scale (IaaS)

Platforms manage these building blocks

- Provision and connect VMs or containers automatically
- Handle networking, storage, and scaling under the hood
- Hide hardware complexity from the developer

Two families:

- **Cloud platforms** (AWS, Azure, GCP, OpenStack): full infrastructure stack
- **Container orchestration** (Kubernetes): deploy and scale containerized workloads

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Platform	Description	Logo
OpenStack	Open-source, widely used in private clouds	 openstack.
VMware vSphere	Enterprise leader, proprietary	
Proxmox VE	Open-source, gaining traction since recent VMware price rises	

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Amazon Web Services (AWS)	Market leader since 2006	
Microsoft Azure	Strong enterprise integration	
Google Cloud Platform (GCP)	Data and AI focus	

These map directly to the IaaS model we just covered: the provider (or you) manages compute, storage, and networking.

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Cooling / Power	Provider	You

Public cloud trades control for simplicity; private cloud gives full control at the cost of operational overhead.

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Hosted / Public: same idea, managed by the cloud provider.

Examples: AWS ECS/EKS, Azure AKS, Google Kubernetes Engine (GKE), OpenShift on major clouds.

Discussion: Platforms

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- What if traffic spikes overnight?

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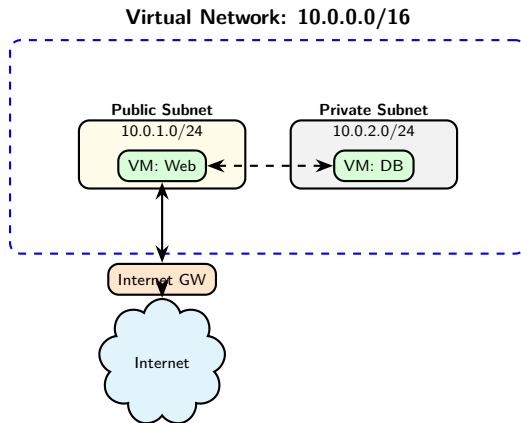
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Key: under the hood, cloud providers use **overlay networks** (covered in Block 5) to isolate tenants at scale.

Virtual Network Architecture



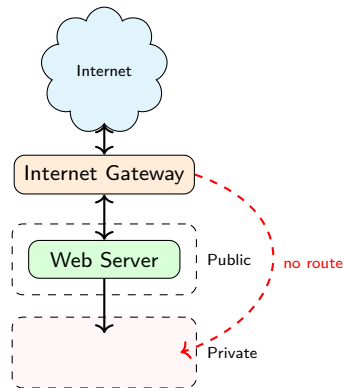
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Public subnet:

- Route to an **Internet Gateway**

Private subnet:



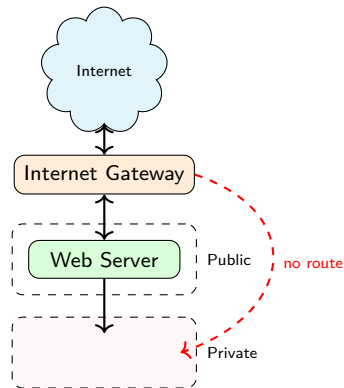
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- Instances can have **public IPs**

Private subnet:



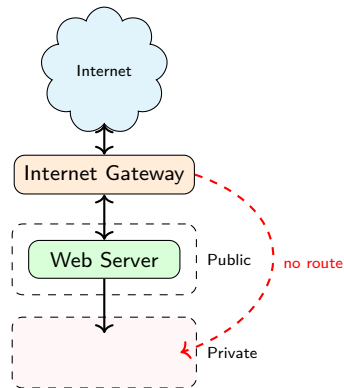
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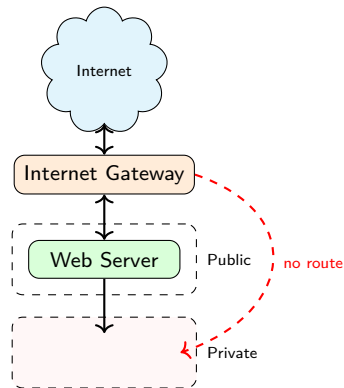
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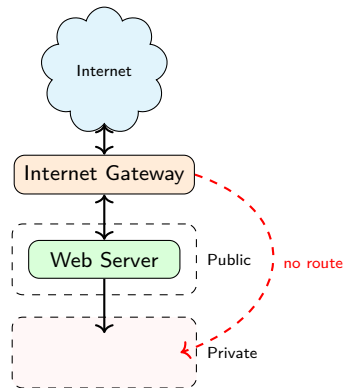
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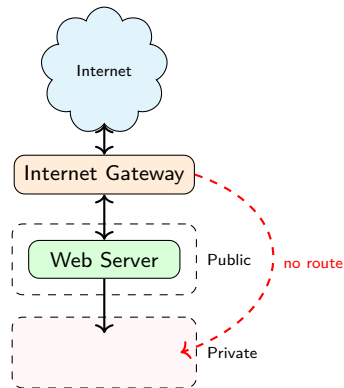
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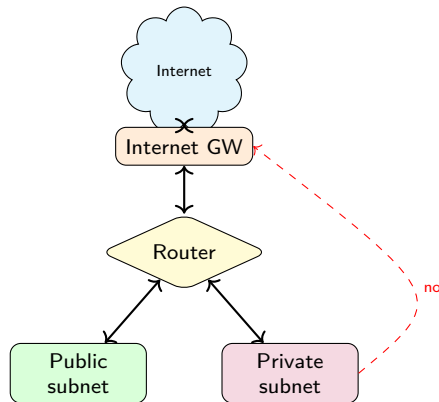


Route Tables

Every virtual network has an implicit **virtual router** managed by the provider. Route tables tell it where to forward traffic.

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0.0.0.0/0	Internet GW	Public only

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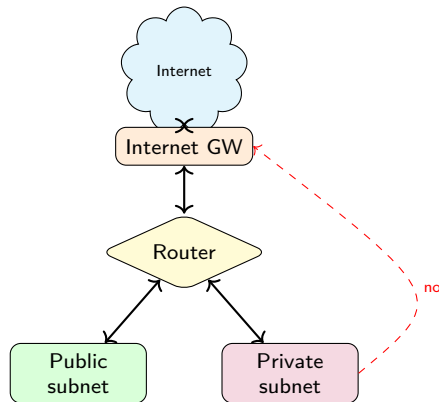


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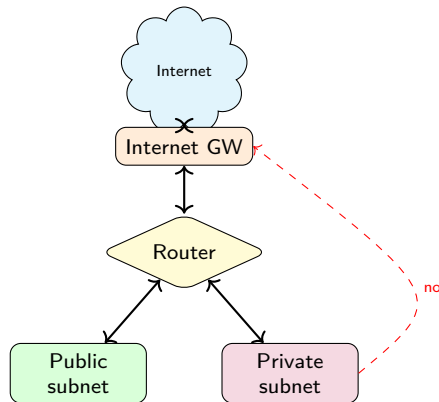


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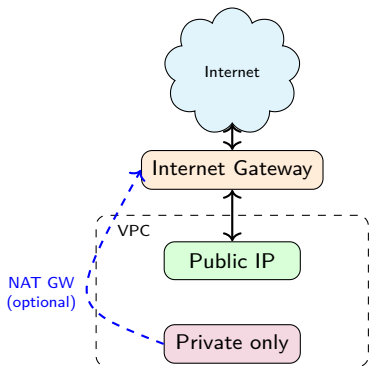
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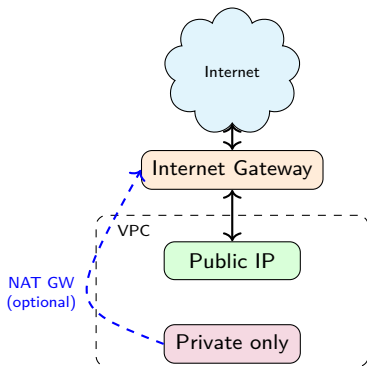
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In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside

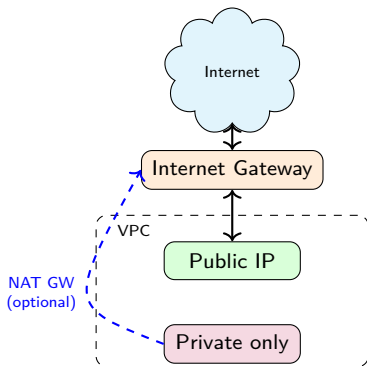
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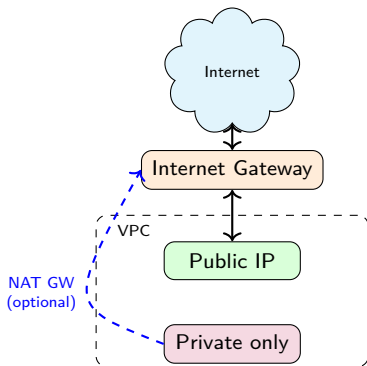
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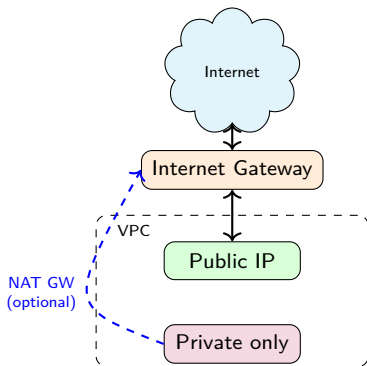
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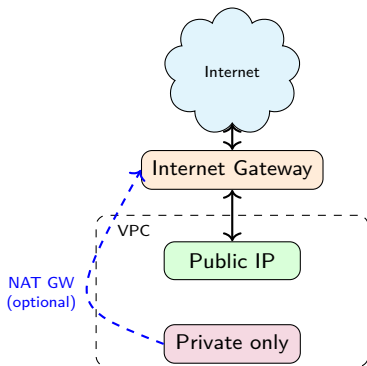
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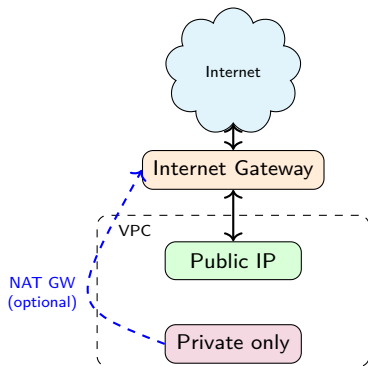
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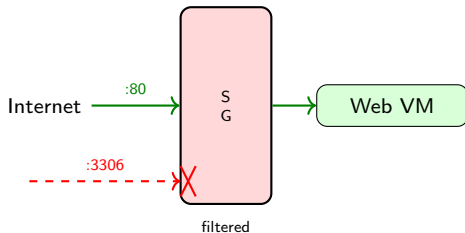
Note: a NAT Gateway can be added for private instances that need outbound Internet access (e.g. updates), without exposing them inbound.

Security Groups: Instance-Level Firewall

Once traffic reaches an instance, you still need to control what it can send and receive.

- **Security group** = virtual firewall on an instance
- Specifies allowed **inbound/outbound** traffic
- **Stateful**: allow inbound → response automatically allowed

Dir.	Proto	Port	Src/Dst
In	TCP	80	0.0.0.0/0
In	TCP	443	0.0.0.0/0
In	TCP	22	10.0.0.0/16
Out	All	All	0.0.0.0/0



By default: all inbound denied, all outbound allowed.

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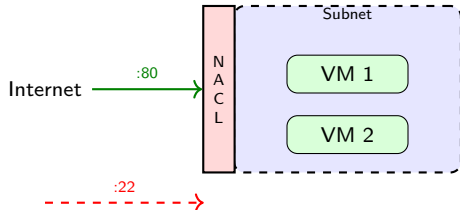
³AWS, "Security Groups for Your VPC," docs.aws.amazon.com.

Network ACLs (Access Control Lists): Subnet-Level Firewall

Network ACLs add a layer of control at the **subnet boundary**, before traffic reaches any instance.

- **Network ACL** = firewall at the **subnet** level
- Rules evaluated in **order** (numbered, first match wins)
- **Stateless**: must explicitly allow inbound AND outbound

#	Dir.	Proto	Port	Action
100	In	TCP	80	ALLOW
200	In	TCP	443	ALLOW
*	In	All	All	DENY



Filter applies to **all** instances in the subnet.

4

⁴AWS, "Network ACLs," docs.aws.amazon.com.

Security Groups vs Network ACLs

	Security Groups	Network ACLs
Scope	Instance level	Subnet level
State	Stateful	Stateless
Rules	Allow only	Allow + Deny
Evaluation	All rules evaluated	First match wins
Default	Deny all inbound	Allow all

Defense in depth: use both together.

- Security Groups → fine-grained per-instance control
- Network ACLs → broad subnet-level guardrails

Common mistake

ACLs are **stateless**: if you allow inbound TCP port 80, you must *also* explicitly allow outbound ephemeral ports (1024–65535) for the response. Forgetting this blocks return traffic.

Discussion: Cloud Networking

*We just deployed CloudBite on AWS.
Our web server must be reachable from
users, but the database must not be
exposed to the Internet. How do we set
this up?*

Think about:

- Our web server needs a public subnet with an Internet Gateway route
- Our database goes in a private subnet: no direct route to the Internet
- Security group on the DB: allow traffic only from our web server
- NAT Gateway if the DB needs outbound access (e.g. pulling updates)

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Three services solve these problems: load balancers, proxies, and VPNs.

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

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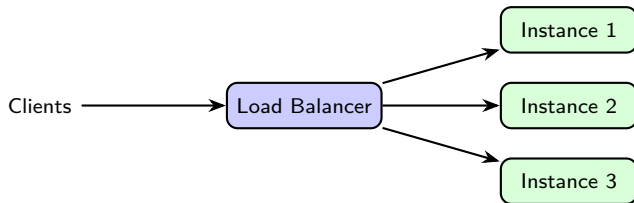
These services run as **managed offerings**: no servers to install or maintain.

Note: Proxy, Reverse Proxy, and VPN will be covered in depth in Block 6. Here we introduce them briefly.

Private cloud: these services are **not** provided automatically; you must deploy and manage them yourself (e.g. HAProxy for load balancing, OpenVPN for VPN).

Load Balancer

- Distributes incoming traffic across **multiple instances** (VMs or containers)
- Prevents any single instance from being overwhelmed
- If one instance fails, traffic is **redirected** to healthy ones

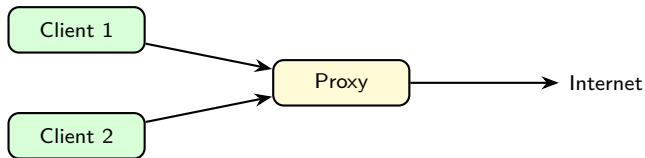


- **L4 (Transport Layer)** Load Balancer: routes based on IP and port (fast, simple)
- **L7 (Application Layer)** Load Balancer: routes based on HTTP headers, URL path, cookies (smarter)
- Cloud examples: AWS ALB/NLB, Azure Load Balancer, GCP Cloud Load Balancing

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

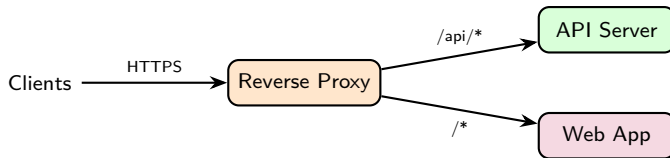
- Clients send requests to the proxy, which forwards them to the Internet



- **Access control:** block certain websites or domains
 - Security: block known malicious domains
 - Company policy: restrict access to non-work-related content
- **Caching:** store frequently accessed content, reduce bandwidth
- **Logging:** monitor what employees or VMs access
- Examples: Squid, corporate firewalls with proxy mode

Reverse Proxy

A reverse proxy sits in front of backend servers and mediates all inbound traffic. Clients never communicate directly with the backend.

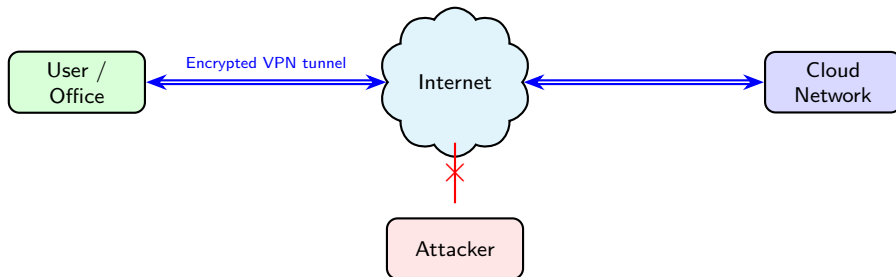


- **TLS termination:** handles HTTPS, backends receive plain HTTP
- **URL routing:** directs requests to the right backend service
- **Hides backend servers:** clients only see the proxy's address
- Examples: NGINX, HAProxy, AWS CloudFront, Azure Front Door

Will be covered in depth in Block 6.

VPN: Virtual Private Network

- A **VPN (Virtual Private Network)** creates a **secure, encrypted tunnel** over the public Internet
- Connects remote networks or users as if they were on the same private network



Two main types:

- **Site-to-site VPN:** office ↔ cloud network
- **Client VPN:** one user ↔ remote network
- Examples: AWS VPN, Azure VPN Gateway, GCP Cloud VPN

Discussion: Cloud Network Services

A company has a public frontend, a private API, and a database. Employees also need to access the cloud from the office. Which services would you use?

Think about:

- Load balancer in front of frontend (public)
- Reverse proxy to route `/api/*` to API server
- Database in private subnet, no public IP
- Site-to-site VPN for office access
- Security groups + ACLs at each tier

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CloudBite, Five Years Later

- We grew to 2 million users. AWS bill: **\$40,000/month** and rising
- Egress fees alone: \$5,000/month (data leaving AWS to users)
- The team now has 30 engineers; we *could* run our own servers
- We now ask: “Should we leave the cloud?”

The cloud is not always the cheapest option. Let us explore why.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on AWS for 15 years
- In 2023, they **moved everything to their own hardware**: no new staff needed
- Result: **\$10 million saved over five years** (50%+ cost reduction)

⁶37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

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Why it worked for them:

- **Stable, predictable workload**: no sudden traffic spikes
- **Large, experienced ops team**: already managing servers regardless
- **No need for global presence**: most users in the same region

Key insight

Basecamp's exit worked because their workload was **predictable**. For a startup with unpredictable traffic, the cloud's elasticity remains essential.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

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 - AWS charges \$0.09/GB for outbound traffic
 - Moving 50 TB out of AWS $\approx$ \$4,500 in fees alone
 - Egress fees increased 20% in 2023; inter-Availability-Zone (AZ) fees doubled in 2025

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- **68% of enterprises** exceeded their cloud budget due to unexpected data transfer costs (Flexera, 2024)

7

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The Pricing Ratchet

- Cloud providers attract customers with **low initial pricing and free tiers**
- Once adoption deepens, prices increase:
 - AWS EC2 instance families: significant price rises in 2025
 - Azure: 5%+ on subscriptions, 10% on Premium SSDs (2025)
 - Google Workspace: 20%–34% increase (March 2025)

⁸InfotechLead, "AWS, Azure, and Google Cloud adjust billing models in 2025," 2025.

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- This pattern appears across industries:
 - **Video streaming**: low subscription prices to gain users, then steady increases once the audience is locked in
 - **Cloud computing**: same dynamic, but switching costs are higher because your entire architecture depends on the provider

The pattern

Low prices → adoption → dependency → price increases → switching is too expensive.

So... Cloud or Not?

	Cloud (public)	On-premises / private
Best when	Unpredictable traffic, small team	Stable workload, large ops team
Cost model	OpEx (pay-as-you-go)	CapEx (buy hardware upfront)
Scaling	Instant, global	Weeks to months (buy + install)
Risk	Vendor lock-in, price hikes	Hardware failures, staffing
Control	Limited (provider's rules)	Full (your hardware, your rules)

- **Most real-world deployments are hybrid:** critical/stable workloads on-premises, burst/variable workloads in the cloud
- There is no universal answer; **every case is a trade-off**

Discussion: The Cloud Trade-Off

CloudBite now has 30 engineers, 2M users, and stable traffic. Should we leave the cloud, go hybrid, or stay? Why?

Think about:

- Is the workload predictable or bursty?
- Does the team have ops expertise?
- How much vendor-specific code exists (Lambda, DynamoDB)?
- What would migration cost in time and money?
- Could a hybrid approach capture the best of both?

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Key Takeaways: CloudBite's Journey

What we built:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Chose a **platform** (AWS, public cloud)
- 5 Created a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added **load balancer**, **reverse proxy**, **VPN**

What we learned:

- 8 The cloud is **not free**: costs grow with scale
- 9 **Vendor lock-in** makes leaving expensive
- 10 Prices increase once you depend on the provider
- 11 On-premises works for **stable, predictable** workloads
- 12 **Hybrid** is often the pragmatic answer
- 13 **Every case is a trade-off**; there is no universal solution

References

- ❶ NIST SP 800-145, Mell & Grance, “The NIST Definition of Cloud Computing,” 2011.
- ❷ AWS, “Amazon VPC User Guide,” docs.aws.amazon.com.
- ❸ Microsoft Azure, “Azure Virtual Network Documentation,” docs.microsoft.com.
- ❹ Google Cloud, “VPC Documentation,” cloud.google.com/vpc/docs.
- ❺ CSA, “Security Guidance for Critical Areas of Focus in Cloud Computing v5,” 2022.
- ❻ Erl, Puttini & Mahmood, *Cloud Computing: Concepts, Technology & Architecture*, Prentice Hall, 2013.
- ❼ Kurose & Ross, *Computer Networking: A Top-Down Approach*, 8th ed., Pearson, 2021.
- ❽ AWS, “Elastic Load Balancing Documentation,” docs.aws.amazon.com.
- ❾ NGINX, Inc., “What Is a Reverse Proxy?,” nginx.com/resources.
- ❿ AWS, “AWS VPN Documentation,” docs.aws.amazon.com.